Procedure

1. Open Android Studio IDE → go to File → New → New Project → specify the application name "CanvaAnimation" and company domain "com.mad.canani" → click "next" → choose Minimum SDK "API 17:Android 4.2(Jelly Bean)" → click "Next" → choose "Blank Activity" → click "next" → specify the Activity Name "MainActivity" → click "Finish".
2. Open MainActivity.java under app/java/ canani.mad.com.canvaanimation and type the following codes:

MainActivity.java

```
package canani.mad.com.canvaanimation;

import android.animation.AnimatorSet;
import android.animation.ArgbEvaluator;
import android.animation.ValueAnimator;
import android.app.Activity;
import android.graphics.Color;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;

public class MainActivity extends Activity {

    AnimatedCircleView circleView;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        circleView = (AnimatedCircleView) findViewById(R.id.circleView);
        Button btnAnimate = (Button) findViewById(R.id.btnAnimate);

        btnAnimate.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                startAnimation();
            }
        });
    }

    private void startAnimation() {

        // Radius animation
        ValueAnimator radiusAnimator = ValueAnimator.ofFloat(50f, 150f);
        radiusAnimator.setDuration(1000);
        radiusAnimator.addUpdateListener(new ValueAnimator.AnimatorUpdateListener() {
            @Override
```

```
            public void onAnimationUpdate(ValueAnimator animation) {
                float value = (float) animation.getAnimatedValue();
                circleView.setRadius(value);
            }
        });

        // Color animation
        ValueAnimator colorAnimator = ValueAnimator.ofObject(
                new ArgbEvaluator(),
                Color.BLUE,
                Color.RED
        );
        colorAnimator.setDuration(1000);
        colorAnimator.addUpdateListener(new ValueAnimator.AnimatorUpdateListener() {
            @Override
            public void onAnimationUpdate(ValueAnimator animation) {
                int color = (int) animation.getAnimatedValue();
                circleView.setCircleColor(color);
            }
        });

        // Play animations together
        AnimatorSet set = new AnimatorSet();
        set.playTogether(radiusAnimator, colorAnimator);
        set.start();
    }
}
```

3. Right click on canani.mad.com.canvaanimation package → New → Java Class → specify the class name as AnimatedCircleView → ok.
4.
5. Open AnimatedCircleView.java under app/java/ canani.mad.com.canvaanimation and type the following codes:

AnimatedCircleView.java

```
package canani.mad.com.canvaanimation;

import android.content.Context;
import android.graphics.Canvas;
import android.graphics.Color;
import android.graphics.Paint;
import android.util.AttributeSet;
import android.view.View;

public class AnimatedCircleView extends View {

    private Paint paint;
    private float radius = 50f;
```

```
    public AnimatedCircleView(Context context) {
        super(context);
        init();
    }

    public AnimatedCircleView(Context context, AttributeSet attrs) {
        super(context, attrs);
        init();
    }

    public AnimatedCircleView(Context context, AttributeSet attrs, int defStyleAttr) {
        super(context, attrs, defStyleAttr);
        init();
    }

    private void init() {
        paint = new Paint(Paint.ANTI_ALIAS_FLAG);
        paint.setColor(Color.BLUE);
    }

    @Override
    protected void onDraw(Canvas canvas) {
        super.onDraw(canvas);

        float cx = getWidth() / 2f;
        float cy = getHeight() / 2f;

        canvas.drawCircle(cx, cy, radius, paint);
    }

    public void setRadius(float radius) {
        this.radius = radius;
        invalidate(); // redraw view
    }

    public void setCircleColor(int color) {
        paint.setColor(color);
        invalidate();
    }
}
```

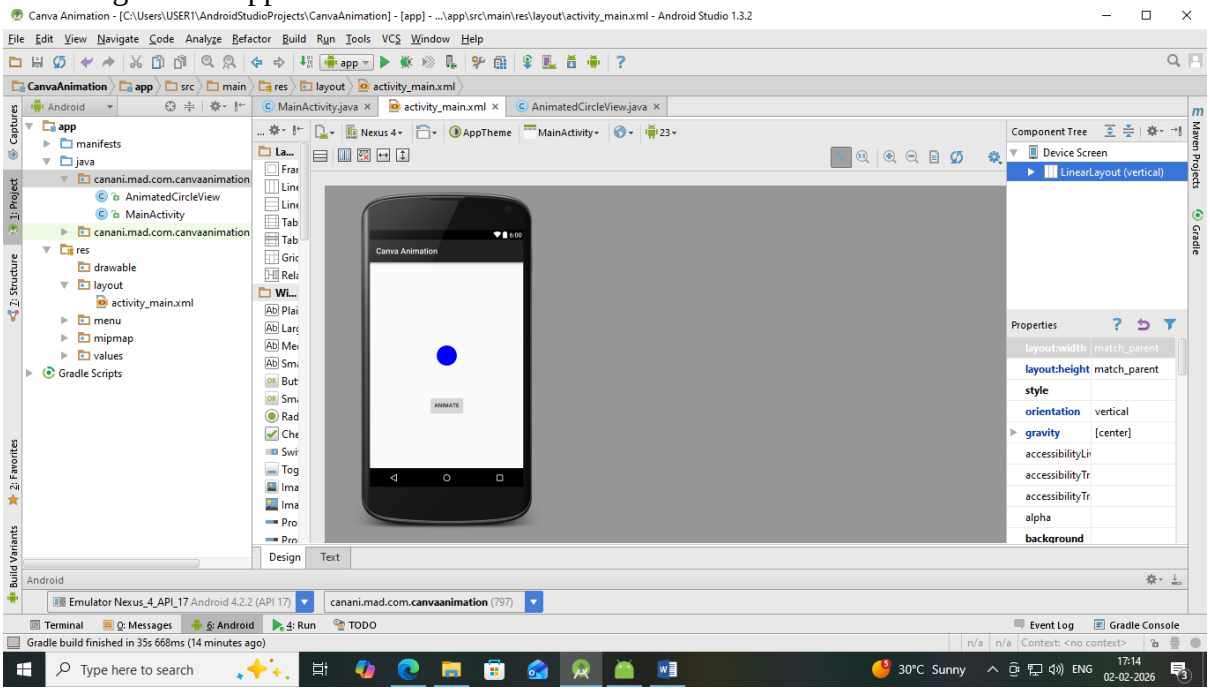
6. Open activity_main.xml under app/res/layout and type the following codes:

activity_main.xml

```
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="center"
```

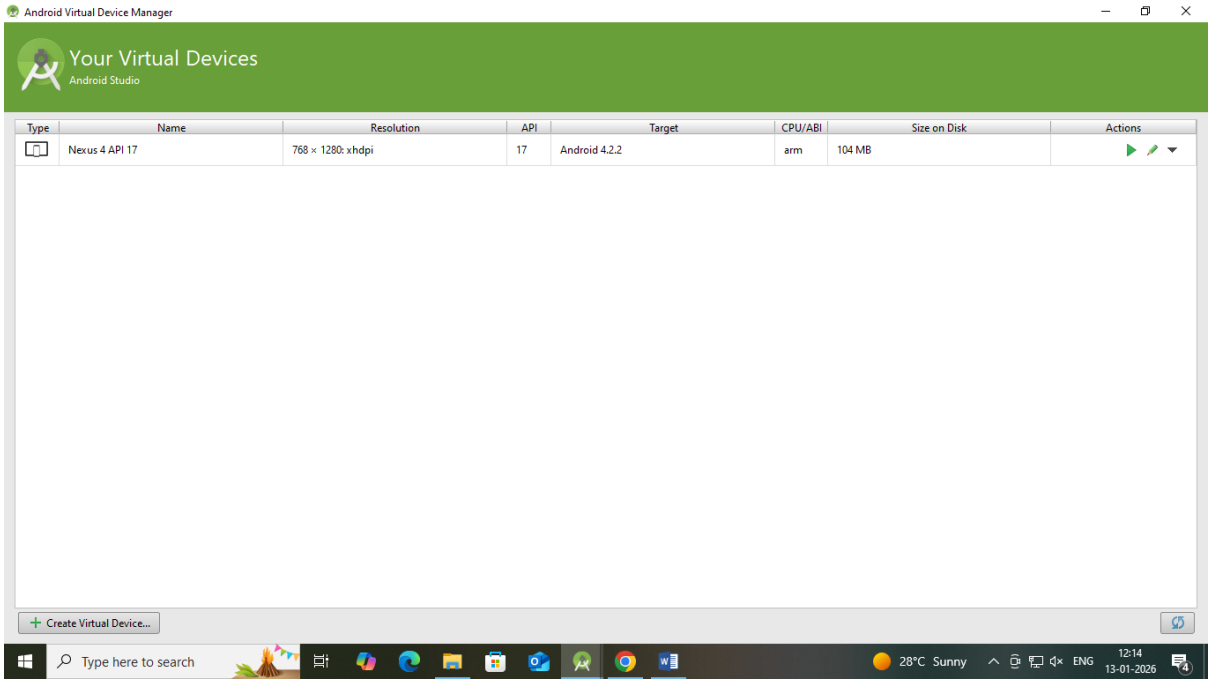
```
    android:orientation="vertical">

    <canani.mad.com.canvaanimation.AnimatedCircleView
        android:id="@+id/circleView"
        android:layout_width="200dp"
        android:layout_height="200dp" />

    <Button
        android:id="@+id/btnAnimate"
        android:text="Animate"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"/>
</LinearLayout>
```

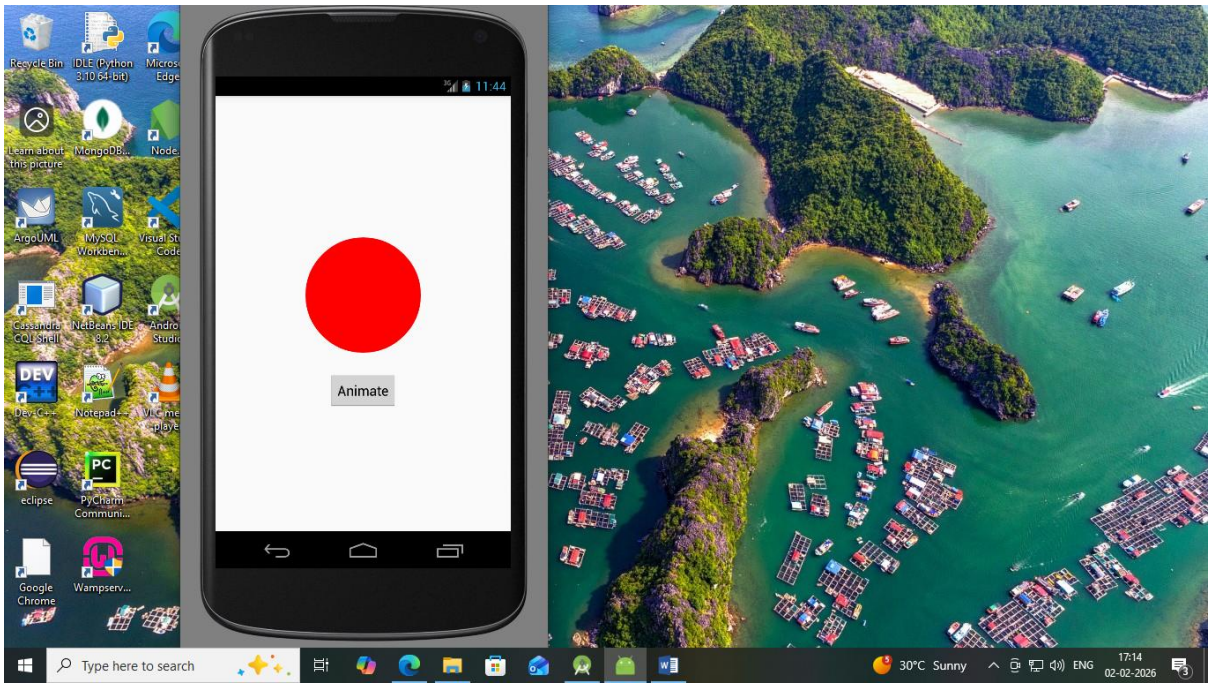
7. The design of the application will be as follows:



8. Go to Tools → android → AVD Manager → click "+ create a virtual device" → select "phone" from category → select "Nexus 4" from the list → click "next" → select Release name: Jelly Bean, API Level: 17, ABI: armeabi-v7a, Target: Android 4.2.2 from the list → click "next" → Choose orientation "portrait" → click "finish".

The following window will appear after configuring AVD:



9. Click “Run app” button in the Android Studio → choose android virtual device → click “ok”.

Output